

*"99.9% of science is boring."*  
*- Makise Kurisu*

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# PRECALCULUS

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August 6, 2026

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# 1 FUNCTIONS

## 1.1 Intro To Functions

**Definition 1.1 (Function).** A function is a relationship that maps 1 input to 1 output.

**Note:** An input never gives more than 1 output.

A function usually looks like

$$y = f(x)$$

$x$  is the *independent* variable.

$y$  or  $f(x)$  is the *Dependent* variable.

Where the standard ordered pairs looks like

$$(x, y) \vee (x, f(x))$$

**Definition 1.2 (Domain).** A domain is a set of inputs.

**Definition 1.3 (Range).** A range is a set of outputs.

**Example 1.1 (Real Example).**

| Work Hours | Pay   |
|------------|-------|
| 25         | \$500 |
| 53         | \$310 |
| 30         | \$490 |
| 40         | \$490 |

**Note:** This is a function but not a *one-to-one function*.

Domain : 23, 53, 30, 40

Range : \$500, \$310, \$490

**Example 1.2 (Input Output).**

$$S = (-2, 16), (-1, 4), (0, 3), (1, 4)$$

This is a function.

$$R = (-2, 5), (3, 9), (5, 0), (-2, 6)$$

This is not a function.

**Example 1.3 (Algebraic).**

$$y = \frac{1}{2}x - 2$$

This is a function.

$$y = 2x^2 - 5x + 4$$

This is a function.

$$y = \pm\sqrt{3-2x}$$

This is not a function because we get more than 1 output for only 1 input.

**Note:**  $y^2 = x \neq y = \sqrt{x}$ , instead it equals to  $y = \pm\sqrt{x}$ . Which makes it not a function.

**Example 1.4 (Factoring).**

$$xy + 2y = 3x - 1$$

$$y(x + 2) = 3x - 1$$

$$y = \frac{3x - 1}{x + 2}$$

## 1.2 Evaluating Functions

**Note:**

$$y = -7x + 5 \equiv f(x) - 7x + 5$$

$$y = \frac{3}{2}x - 1 \equiv g(x) = \frac{3}{2}x - 1$$

We use the  $f(x)$  notation to know what the input is.

**Example 1.5 (Exponents).**

$$h(x) = -2x^2 + x - 1$$

1.  $h(-1) = -2(-1)^2 + (-1) - 1$   
 $h(-1) = -4$   
 $(-1, -4)$
2.  $h(-x) = -2(-x)^2 + (-x) - 1$   
 $h(-x) = -2x^2 + x - 1$
3.  $h(x+1) = -2(x+1)^2 + (x+1) - 1$   
 $h(x+1) = -2(x^2 + 2x + 1) + (x+1) - 1$   
 $h(x+1) = -2x^2 - 4x - 2 + x + 1 - 1$   
 $h(x+1) = -2x^2 - 3x - 1$

**Example 1.6 (Square Root).**

$$f(x) = \sqrt{x^2 - 3x}$$

1.  $f(5) = \sqrt{(5)^2 - 3(5)}$   
 $f(5) = \sqrt{10}$   
 $(5, \sqrt{10})$
2.  $f(0) = \sqrt{(0)^2 - 3(0)}$   
 $f(0) = \sqrt{0}$   
 $(0, 0)$
3.  $f(2x) = \sqrt{(2x)^2 - 3(2x)}$   
 $f(2x) = \sqrt{4x^2 - 6x}$
4.  $f(x+h) = \sqrt{(x+h)^2 - 3(x+h)}$   
 $f(x+h) = \sqrt{x^2 + 2xh + h^2 - 3x - 3h}$
5.  $f(1) = \sqrt{(1)^2 - 3(1)}$   
 $f(1) = \sqrt{-2}$

**Note:** This is a domain issue so we can't use 1.

**Note:** For domains:

1. **Square Roots:** The inside of the square root (*radicand*) must always always positive.
2. **Denominator:** The denominator must never be 0.

**Example 1.7 (Fractions).**

$$g(x) = \frac{2x+1}{x-5}$$

$$1. \quad g(0) = \frac{2(0)+1}{(0)-5}$$

$$g(0) = \frac{1}{-5}$$

$$2. \quad g(x^2) = \frac{2(x^2)+1}{(x^2)-5}$$

$$g(x^2) = \frac{2x^2+1}{x^2-5}$$

$$3. \quad g(5) = \frac{2(5)+1}{(5)-5}$$

$$g(5) = \frac{11}{0}$$

**Note:** This is also a domain issue.

$$4. \quad -g(x) = -\frac{2x+1}{x-5}$$

### 1.2.1 Difference Quotient

**Definition 1.4 (Difference Quotient).** A formula used in algebra and calculus to find the **average rate of change** of a function over a small interval

$$\frac{f(x+h) - f(x)}{h}$$

**Example 1.8 (Example).**

$$f(x) = x^2 + 1$$

$$f(x+h) = (x+h)^2 + 1$$

$$f(x+h) = x^2 + xh + xh + h^2 + 1$$

$$f(x+h) = x^2 + 2xh + h^2 + 1$$

$$\frac{f(x+h)-f(x)}{h} = \frac{(x^2+2xh+h^2+1)-(x^2+1)}{h}$$

$$\frac{f(x+h)-f(x)}{h} = \frac{2xh+h^2}{h}$$

$$\frac{f(x+h)-f(x)}{h} = \frac{h(2x+h)}{h}$$

$$\frac{f(x+h)-f(x)}{h} = 2x + h$$

### 1.3 Finding Domains

**Definition 1.5 (Domains).** Domains is the set of inputs of a function that give a  $\mathbb{R}$  number output.

#### 1.3.1 The Problem Areas

1. **Square Roots:**  $\sqrt{(\quad)}$  radicand  $> 0$ .
2. **Denominators:**  $\text{denominator} \neq \text{zero}$ .

**Example 1.9 (Square Root).**

$$\begin{aligned} f(x) &= \sqrt{5-4x} \\ 5-4x &\geq 0 \\ -4x &\geq -5 \\ x &\geq \frac{-5}{-4} \\ x &\leq \frac{5}{4} \end{aligned}$$

Can be written as

$$D = \{x \mid x \leq \frac{5}{4}\}$$

Or

$$(-\infty, \frac{5}{4}]$$

**Example 1.10 (Denominator).**

$$\begin{aligned} g(t) &= \frac{5t}{t^3-16t} \\ t^3-16t &= 0 \\ t(t^2-16) &= 0 \\ t(t-4)(t+4) &= 0 \\ t \neq 0 \vee t \neq 4 \vee t \neq -4 \end{aligned}$$

Can be written as

$$D = \{t \in \mathbb{R} \mid t \neq 0, t \neq -4, t \neq 4\}$$

Or

$$(-\infty, -4) \cup (-4, 0) \cup (0, 4) \cup (4, \infty)$$

**Example 1.11 (Examples).**

1.  $f(x) = 3x^2 - 2x + 4$   
 $D : \{x \in \mathbb{R}\} \text{ Or } (-\infty, \infty)$



2.  $h(x) = \frac{\sqrt{3x-15}}{x-4}$

$$x \neq 4$$

$$3x - 15 \geq 0$$

$$3x \geq 15$$

$$x \geq 5$$

$$D : \{x \mid x \neq 4\} \text{ or } [5, \infty)$$

3.  $g(x) = \frac{5}{\sqrt{x-8}}$

$$x - 8 > 0$$

$$D : \{x \mid x > 8\} \text{ or } (8, \infty)$$

## 1.4 Operations Of Functions

**Definition 1.6 (Operations Of Function).** Function operations involve combining two or more functions to create a new function using standard arithmetic: **addition, subtraction, multiplication and division.**

**Example 1.12 (Operations).**

$$f(x) = \frac{2x+3}{3x-2} \quad g(x) = \frac{4x}{3x-2}$$

$$D : \left\{ x \mid x \neq \frac{2}{3} \right\}$$

1.

$$(f+g)(x) = \frac{2x+3}{3x-2} + \frac{4x}{3x-2} \rightarrow \frac{6x+3}{3x-2} \quad D : \left\{ x \mid x \neq \frac{2}{3} \right\}$$

2.

$$(f-g)(x) = \frac{2x+3}{3x-2} - \frac{4x}{3x-2} \rightarrow \frac{-2x+3}{3x-2} \quad D : \left\{ x \mid x \neq \frac{2}{3} \right\}$$

3.

$$(f \cdot g)(x) = \left( \frac{2x+3}{3x-2} \right) \left( \frac{4x}{3x-2} \right) \rightarrow \frac{8x^2+12x}{(3x-2)(3x-2)} \rightarrow \frac{8x^2+12x}{(3x-2)^2} \quad D : \left\{ x \mid x \neq \frac{2}{3} \right\}$$

4.

$$\left( \frac{f}{g} \right)(x) = \frac{\frac{2x+3}{3x-2}}{\frac{4x}{3x-2}} \rightarrow \frac{2x+3}{3x-2} \cdot \frac{3x-2}{4x} \rightarrow \frac{2x+3}{4x} \quad D : \left\{ x \mid x \neq \frac{2}{3}, x \neq 0 \right\}$$

**Example 1.13 (Square Root and Fractions).**

$$f(x) = \sqrt{x+3} \quad g(x) = \frac{5}{x}$$

$$D : \{x \mid x \geq -3\} \quad D : \{x \mid x \neq 0\}$$

1.

$$(f+g)(x) = \sqrt{x+3} + \frac{5}{x} \quad D : \{x \mid x \geq -3, x \neq 0\}$$

2.

$$(f-g)(x) = \sqrt{x+3} - \frac{5}{x} \quad D : \{x \mid x \geq -3, x \neq 0\}$$

3.

$$(f \cdot g)(x) = \frac{5\sqrt{x+3}}{x} \quad D : \{x \mid x \geq -3, x \neq 0\}$$

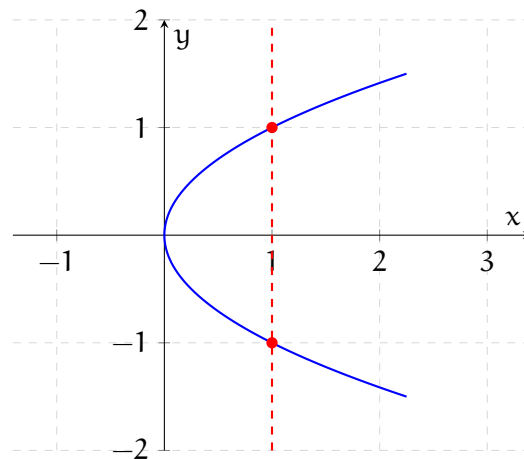
4.

$$\left( \frac{f}{g} \right)(x) = \sqrt{x+3} \cdot \frac{x}{5} \rightarrow \frac{x\sqrt{x+3}}{5} \quad D : \{x \mid x \geq -3, x \neq 0\}$$

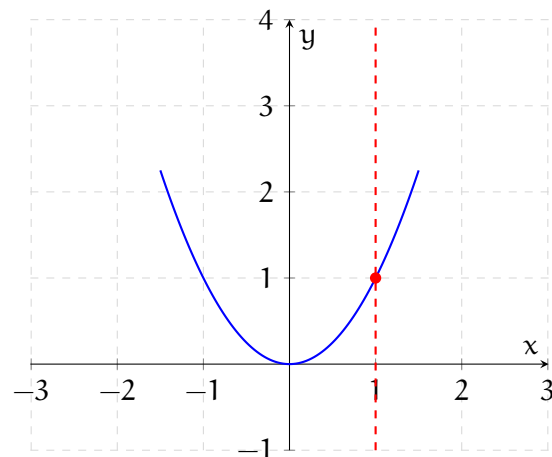
## 1.5 The Vertical Line Test

**Definition 1.7 (Vertical Line Test).** This is a test to determine if a graph is a function or not.

**Example 1.14 (Non Function).** This is not a function because the vertical line passes through 2 points, which means that it outputs more than 1 output.



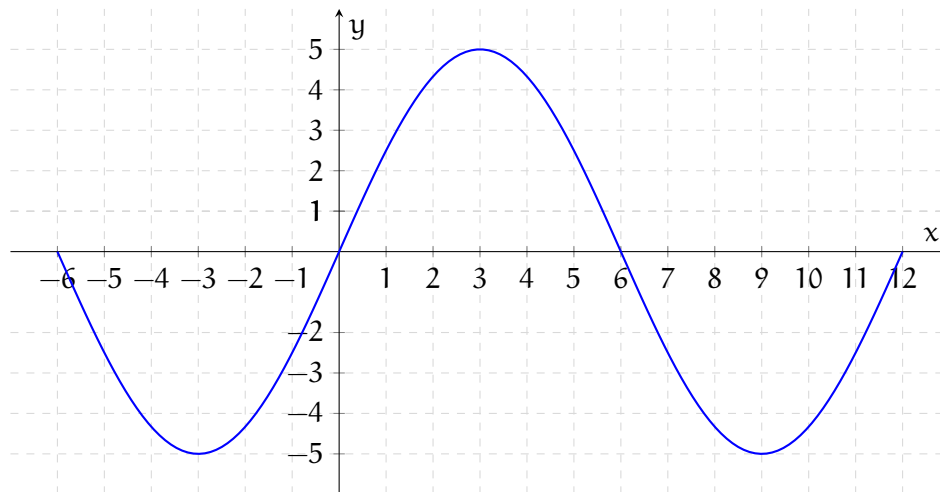
**Example 1.15 (Function).** This is a function because the vertical line passes through only 1 point, which means that it only gives a single output.



## 1.6 Features of Graphs Points, Domain, Range

**Definition 1.8 (Graph Points).** A graph point of a function is an ordered pair  $(x, y)$  on a coordinate plane, where  $y$  is the output generated when the function precesses the input  $x$ .

**Example 1.16 (Graphs).**



1. Find  $f(0) = 0$   $(0, 0)$
2. Find  $f(-2) = -2$   $(-2, -2)$
3. Is  $f(1)$  positive  $\vee$  negative = positive
4. Is  $f(-2)$  positive  $\vee$  negative = negative
5. For what  $x$  is  $f(x) = 0$ ?  
 $(-6, 0) (0, 0) (6, 0) (12, 0)$   $x = -6, 0, 6, 12$
6. For what  $x$  is  $f(x) > 0$ ?  
 $0 < x < 6$   $(-4, 6)$
7. For what  $x$  is  $f(x) < 0$ ?  
 $-6 < x < 0, 6 < x < 12$   $(-6, 0) \cup (6, 12)$
8. Domain:  $D : \{x \mid -6 \leq x \leq 12\}$   $[-6, 12]$
9. Range:  $R : \{f(x) \mid -5 \leq f(x) \leq 5\}$   $[-5, 5]$
10.  $x$  Intercept:  $(-6, 0), (0, 0), (6, 0), (12, 0)$
11.  $y$  intercept:  $(0, 0)$
12. How many times does the  $y = 2$  cross the graph? 2
13. For what  $x$  is  $f(x) = 4$ ?  
 $(2, 4) (4, 4)$
14. For what  $x$  is  $f(x) = -5$ ?  
 $(-3, -5) (9, -5)$

**Definition 1.9 (Square Brackets: Inclusive).** “[” or “]” The endpoint number is included in the interval (corresponds to  $\leq \vee \geq$ ).

**Definition 1.10 (Parentheses: Exclusive).** “(“ or “)” The endpoint number is not included in the interval (corresponds to  $< \vee >$ ). It is also always used for infinity ( $\infty$ ).

**Example 1.17 (Algebraic).**

$$f(x) = \frac{x+2}{x-6}$$

1.  $f(1) = \frac{(1)+2}{(1)-6} \rightarrow \frac{3}{5} \quad (1, -\frac{3}{5})$

2.  $f(x) = 2 \dots$  find  $x$

$$2 = \frac{x+2}{x-6}$$

$$2x - 12 = x + 2$$

$$x - 12 = 2$$

$$x = 14$$

$$(14, 2)$$

3. Domain:

$$x - 6 = 0$$

$$x = 6$$

$$D = \{x \mid x \neq 6\}$$

$$(-\infty, 6) \cup (6, \infty)$$

4.  $x$  intercept:  $f(x) = 0$

$$0 = \frac{x+2}{x-6}$$

$$x + 2 = 0$$

$$x = -2$$

$$(-2, 0)$$

5.  $y$  intercept:

$$f(0) = \frac{(0)+2}{(0)-6} = -\frac{1}{3}$$

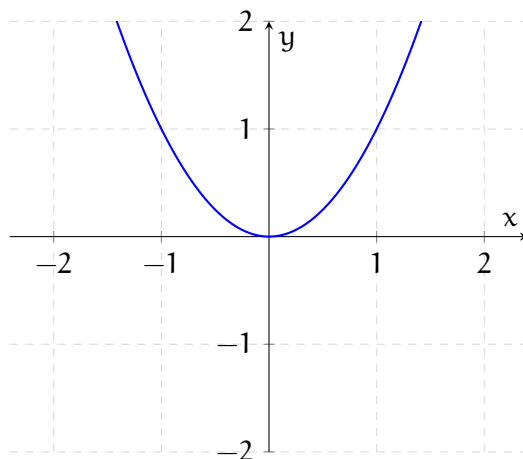
$$(0, -\frac{1}{3})$$

## 1.7 Properties of Functions: Even or Odd

**Definition 1.11 (Even).** Even functions are symmetric about y-axes in a graph.

**Example 1.18 (Even Function).** This is an even function because if we fold it down the y-axis and fold it onto itself, it would be a mirror image perfectly.

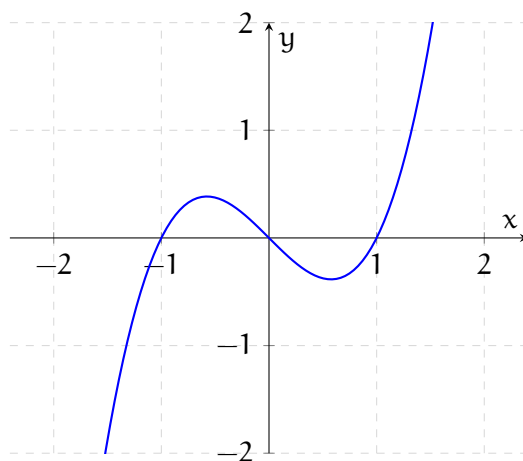
$$f(x) = x^2$$



**Definition 1.12 (Odd).** Odd functions are symmetric about origins in a graph.

**Example 1.19 (Odd Function).** This is an odd function because if we rotate the graph  $180^\circ$  about the origin, it will still be the exact same graph.

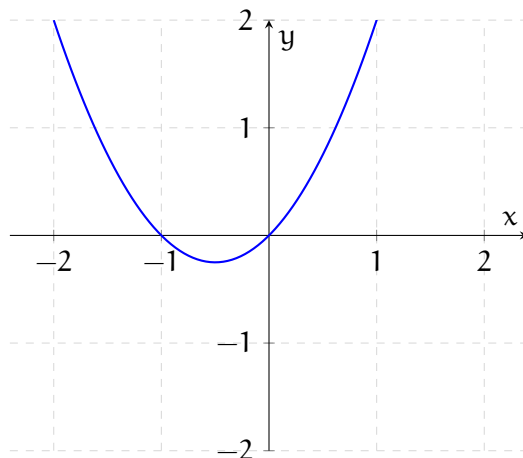
$$f(x) = x^3 - x$$



**Note:** A lot of graphs are neither even or odd. For example:

**Example 1.20 (Neither).**

$$f(x) = x^2 + x$$



**Definition 1.13 (Algebraic Definition of Even Functions).**

$$f(x) = f(-x)$$

**Definition 1.14 (Algebraic Definition of Odd Functions).**

$$-f(x) = f(-x)$$

**Example 1.21 (Algebraic).**

- $f(x) = 2x^4 - x^2$   
 $f(-x) = 2(-x)^4 - (-x)^2$   
 $f(-x) = 2x^4 - x^2$  Even
- $g(x) = \frac{x}{x^2-1}$   
 $g(-x) = \frac{(-x)}{(-x)^2-1} \rightarrow g(-x) = \frac{-x}{x^2-1}$   
 $g(-x) = -\frac{x}{x^2-1}$  Odd
- $h(x) = 3x^3 + 5$   
 $h(-x) = 3(-x)^3 + 5 \rightarrow h(-x) = -3x^3 + 5$   
 $h(-x) = -(3x^3 - 5)$  Neither
- $F(x) = \frac{x^2+3}{x^2-1}$   
 $F(-x) = \frac{(-x)^2+3}{(-x)^2-1}$   
 $F(-x) = \frac{x^2+3}{x^2-1}$  Even
- $G(x) = \sqrt{x}$   
 $G(-x) = \sqrt{(-x)}$  Neither because it's not a  $\mathbb{R}$  number
- $H(x) = \frac{-x^3}{3x^2-9} \rightarrow -\frac{x^3}{3x^2-9}$   
 $H(-x) = \frac{-(-x)^3}{3x^2-9}$   
 $H(-x) = \frac{x^3}{3x^2-9}$  Odd

## 1.8 Properties of Functions: Increasing or Decreasing

### Definition 1.15 (Increasing).

$$\begin{aligned} &\text{for } x_1 < x_2 \\ &f(x_1) < f(x_2) \end{aligned}$$

From left  $\rightarrow$  right in the  $x$ -axis, the graph climbs upward.

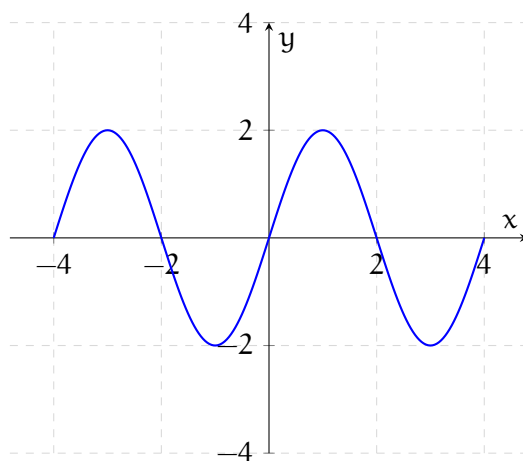
### Definition 1.16 (Decreasing).

$$\begin{aligned} &\text{for } x_1 < x_2 \\ &f(x_1) > f(x_2) \end{aligned}$$

From left  $\rightarrow$  right in the  $x$ -axis, the graph falls downward.

**Note:** All of this is given by an open intervals of the  $x$ -axes.

### Example 1.22 (Decreasing and Increasing Function).

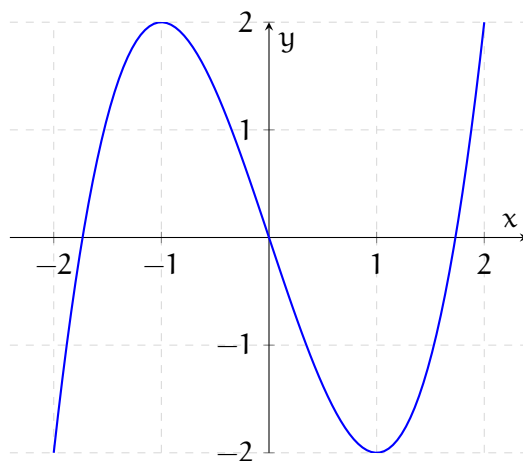


Increasing  $[-1, 1] \cup [3, 4]$

Decreasing  $[-3, -1] \cup [1, 3]$

Another example is:

$$f(x) = -x^3 - 3 * x$$





Increasing  $[-1, 1]$

Decreasing  $(-\infty, -1] \cup [1, \infty)$

## 1.9 Properties of Functions: Extremas

**Definition 1.17 (Local/Relative Maximum).** The highest point in a specific region or a small interval of a function.

$$f(x) \leq f(c) \dots c \text{ is a local max}$$

Where functions changes from increasing to decreasing.

**Definition 1.18 (Local/Relative Minimum).** The lowest point in a specific region or a small interval of a function.

$$f(x) \geq f(c) \dots c \text{ is a local min}$$

Where functions changes from decreasing to increasing.

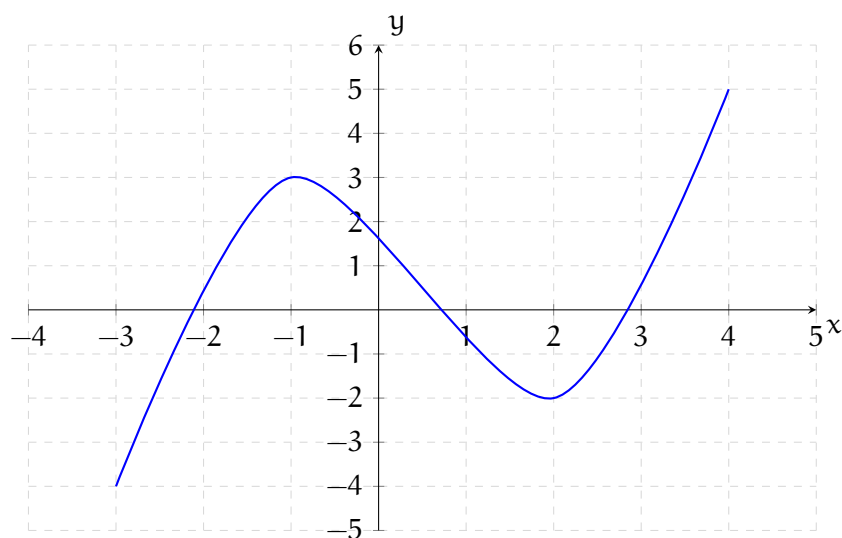
**Note:** Endpoints are not allowed for Local minimum or maximum. But it is allowed for absolute minimum or maximum.

**Definition 1.19 (Absolute Maximum).** Highest output on graph.

**Definition 1.20 (Absolute Minimum).** Lowest output on graph

**Note:** Every continuous closed interval “[,]” will guarantee we have an absolute max and an absolute min.

**Example 1.23 (Graph).**



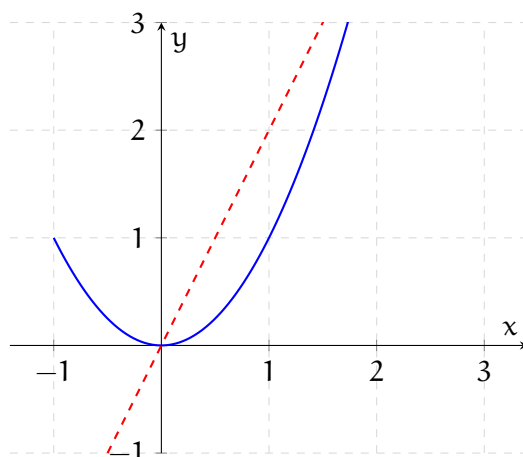
1. Local Minimum:  $-2 @ (2, -2)$
2. Local Maximum:  $3 @ (-1, 3)$
3. Absolute Minimum:  $-4 @ (-3, -4)$
4. Absolute Maximum:  $5 @ (4, 5)$

### 1.10 Average Rate of Change

**Definition 1.21 (Average Rate of Change).** The slope of the line connecting two points on a curve, known as the *secant line*.

$$\text{Average Rate of Change} = \frac{f(b) - f(a)}{b - a}$$

**Example 1.24 (Graph).**



$$a = 0, b = 2$$

$$\frac{f(2) - f(0)}{2 - 0} = \frac{4 - 0}{2 - 0} = 2$$

**Example 1.25 (Algebraic).**

$f(x) = x^2 - 2x$  Find average rate of change

From  $x = 3$  to  $x = 5$

$$f(3) = (3)^2 - 2(3)$$

$$f(3) = 3$$

$$f(5) = (5)^2 - 2(5)$$

$$f(5) = 15$$

$$(3, 3) \quad (5, 15)$$

$$\text{Average Rate of Change} = \frac{15 - 3}{5 - 3}$$

$$\text{Average Rate of Change} = \frac{12}{2} \rightarrow 6$$

### 1.11 Piece-wise Functions

**Definition 1.22 (Piece-wise Functions).** A single function that uses two or more different formulas or rules, with each rule applying to a specific portion of the function's domain.

**Example 1.26 (Algebraic).**

1.

$$f(x) = \begin{cases} x^3 & \text{if } x < 1 \\ 3x - 2 & \text{if } x \geq 1 \end{cases}$$

(a)  $f(-1) = (-1)^3 \rightarrow (-1, -1)$

(b)  $f(0) = (0)^3 \rightarrow (0, 0)$

(c)  $f(1) = 3(1) - 2 \rightarrow (1, 1)$

2.

$$f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 2 & \text{if } x = 0 \\ 2x + 1 & \text{if } x > 0 \end{cases}$$

(a)  $f(-2) = (-2)^2 \rightarrow (-2, 4)$

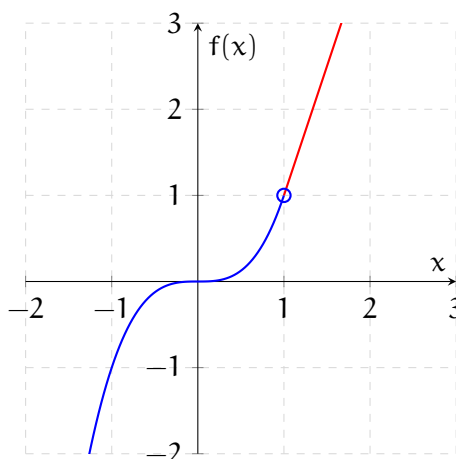
(b)  $f(0) = 2 \rightarrow (0, 2)$

(c)  $f(2) = 2(2) + 1 \rightarrow (2, 5)$

**Example 1.27 (Graph).**

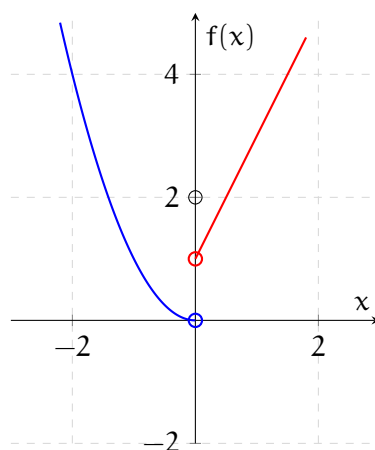
1.

$$f(x) = \begin{cases} x^3 & \text{if } x < 1 \\ 3x - 2 & \text{if } x \geq 1 \end{cases}$$



2.

$$f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 2 & \text{if } x = 0 \\ 2x + 1 & \text{if } x > 0 \end{cases}$$

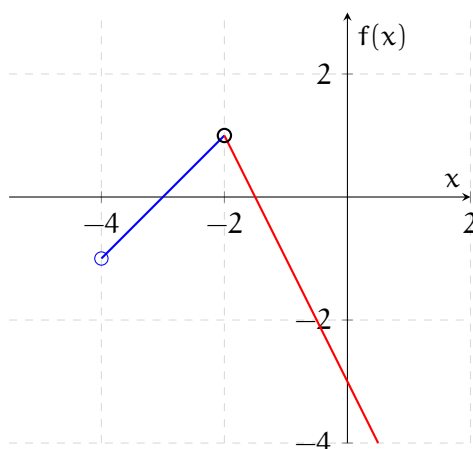


**Definition 1.23 (Not Continuous Function).** In a piece-wise function, a function is not continuous (*discontinuous*) at a point if the different sub-functions do not connect seamlessly at the transition boundary

**Example 1.28 (Not Continuous Function).**

$$f(x) = \begin{cases} x + 3 & \text{if } -4 \leq x < -2 \\ -2x - 3 & \text{if } x > -2 \end{cases}$$

**Example 1.29 (Graph).**



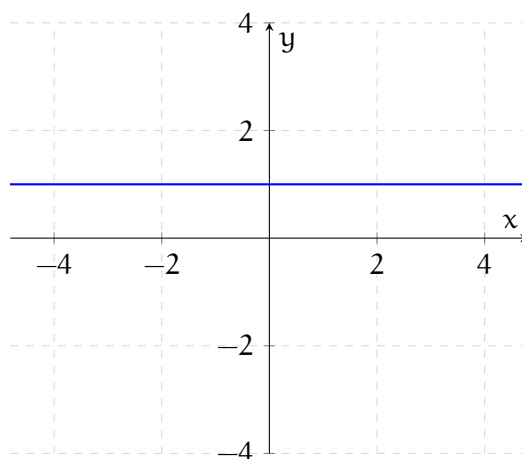
## 1.12 Important Functions

**Example 1.30 (Even).** All of the examples have the point  $(0,0)$   $(1,1)$   $(-1,1)$  except for the  $f(x) = b$  which is just a constant value and the  $f(x) = \sqrt{x}$  function which can't have a negative input ( $x \geq 0$ ).

1.

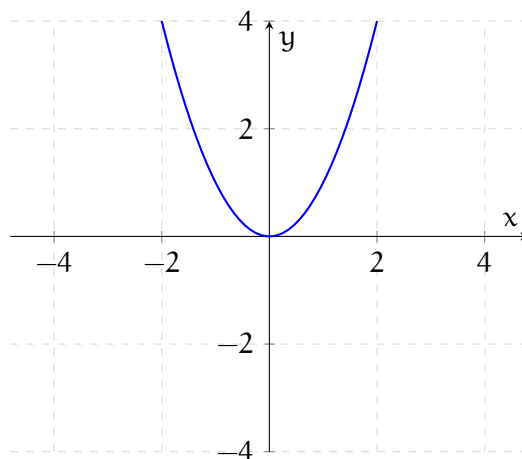
$$f(x) = b$$

For this example  $b = 1$



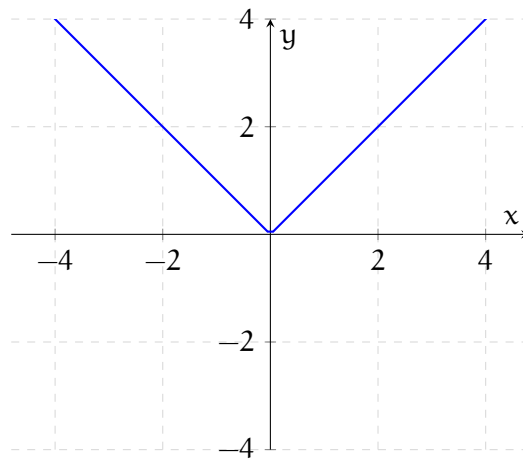
2.

$$f(x) = x^2$$



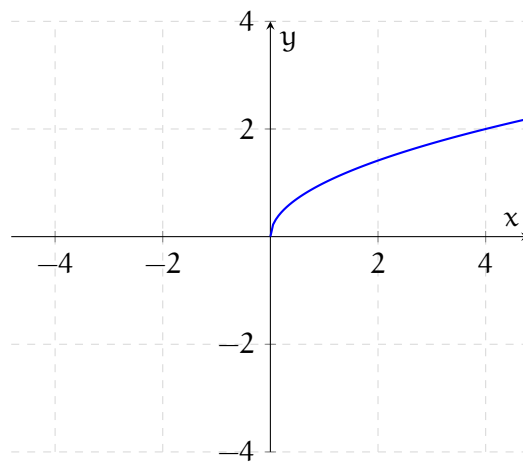
3.

$$f(x) = |x|$$



4.

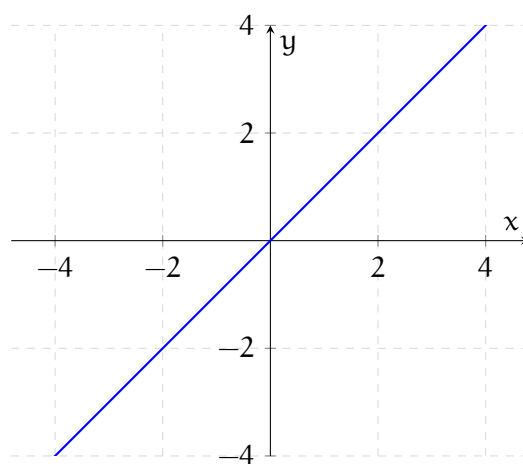
$$f(x) = \sqrt{x}$$



**Example 1.31 (Odd).** All of the examples have the point  $(0,0)$   $(1,1)$   $(-1,-1)$  except for the  $f(x) = \frac{1}{x}$  function which doesn't have the  $(0,0)$  point.

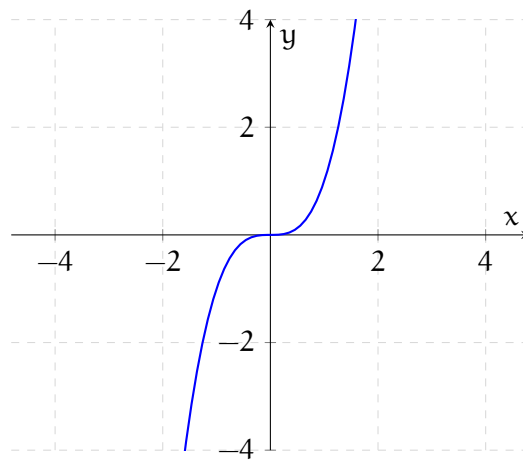
1.

$$f(x) = x$$



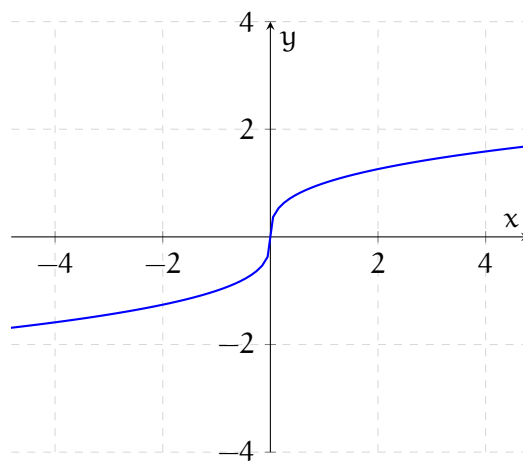
2.

$$f(x) = x^3$$



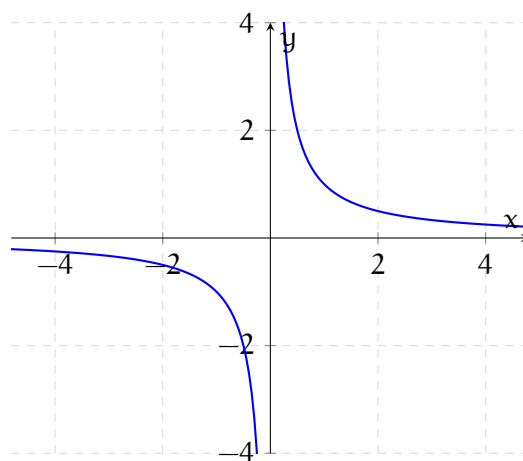
3.

$$f(x) = \sqrt[3]{x}$$



4.

$$f(x) = \frac{1}{x}$$





## 2 GRAPHICAL TRANSFORMATIONS

### 2.1 Introduction To Graphical Transformations

**Definition 2.1 (Graphical Transformations).** A mathematical operation that alters the position, size or orientation of a function's graph on a coordinate plane, without changing its basic shape.

#### 2.1.1 Vertical

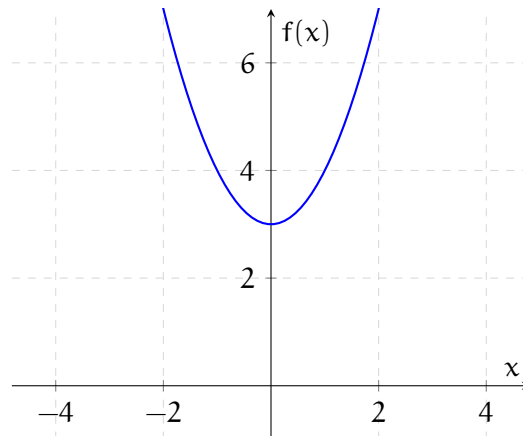
**Definition 2.2 (Vertical Transformation).**

$$f(x) + k \quad \text{Shift Up}$$

$$f(x) - k \quad \text{Shift Down}$$

**Example 2.1 (Vertical Transformation).**

$$f(x) = x^2 + 3$$



**Note:** Key points are  $(1, 1)$ ,  $(0, 0)$ ,  $(-1, 1)$ .

#### 2.1.2 Horizontal

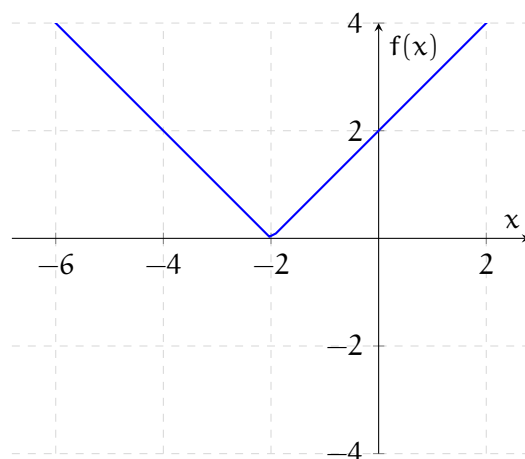
**Definition 2.3 (Horizontal Transformation).**

$$f(x + h) \quad \text{Shift Left}$$

$$f(x - h) \quad \text{Shift Right}$$

**Example 2.2 (Horizontal Transformation).**

$$f(x) = |x + 2|$$



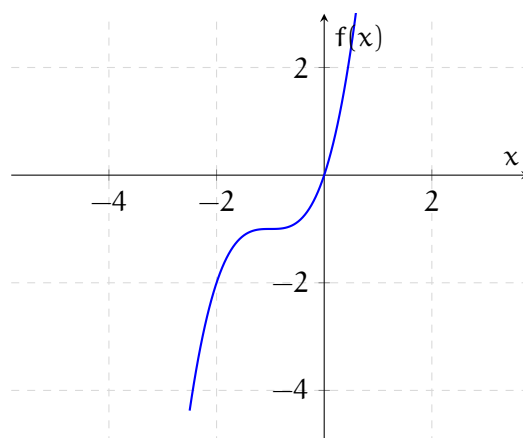
**Note:** Key points are  $(1, 1)$ ,  $(0, 0)$ ,  $(-1, 1)$ .

### Example 2.3 (Examples).

1.

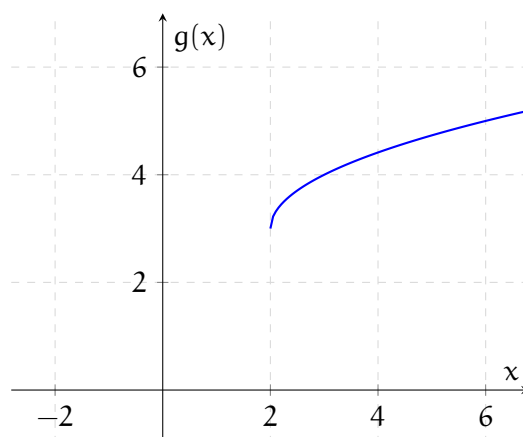
$$f(x) = (x + 1)^3 - 1$$

$$f(x) = (x + 1)^3 - 1$$



2.

$$g(x) = \sqrt{x - 2} + 3$$



## 2.1.3 Vertical Stretch/Compression

**Definition 2.4 (Vertical Stretch/Compression).**

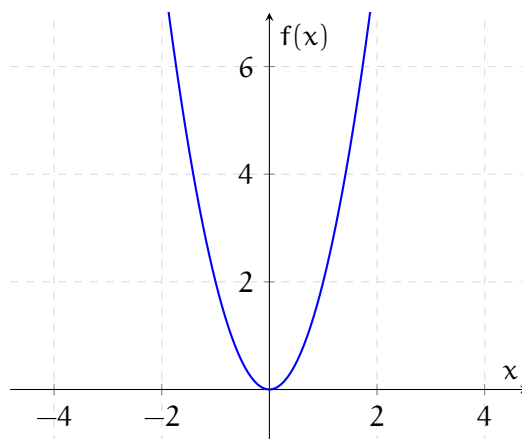
$$y = a f(x)$$

 $a > 1$  Stretch Narrow

 $a < 1$  Compression Wide
**Example 2.4 (Vertical Stretch/Compression).**

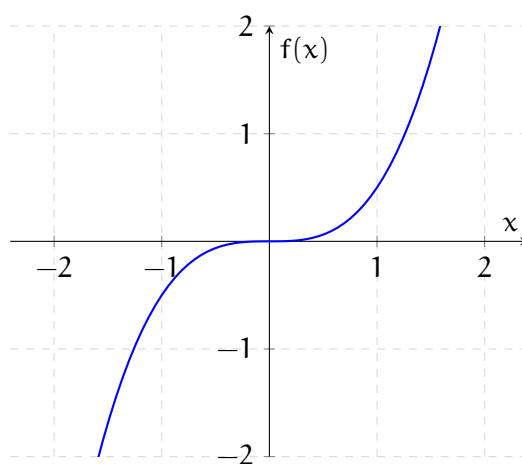
1.

$$f(x) = 2x^2$$

**Note:** We just multiply our outputs by 2.

2.

$$f(x) = \frac{1}{2}x^3$$



## 2.1.4 Horizontal Stretch/Compression

**Definition 2.5 (Horizontal Stretch/Compression).**

$$y = f(ax)$$

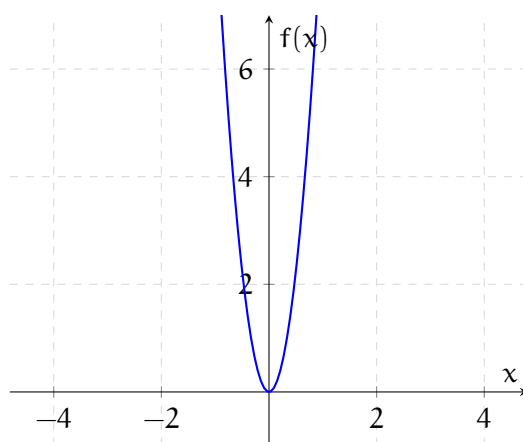
$a > 1$  Horizontal Compression

$a < 1$  Horizontal Stretch

**Note:** Most of the time we can change horizontal stretch/compressions into vertical stretch/compressions respectively.

**Example 2.5 (Horizontal Stretch/Compression).**

$$f(x) = (3x)^2 \rightarrow 9x^2$$



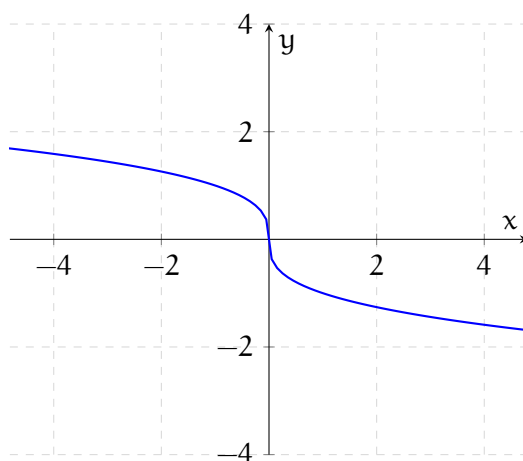
### 2.1.5 Reflection

**Definition 2.6 (Reflection About x Axis).**

$$y = -f(x)$$

**Example 2.6 (Reflection About x Axis).**

$$f(x) = -\sqrt[3]{x}$$

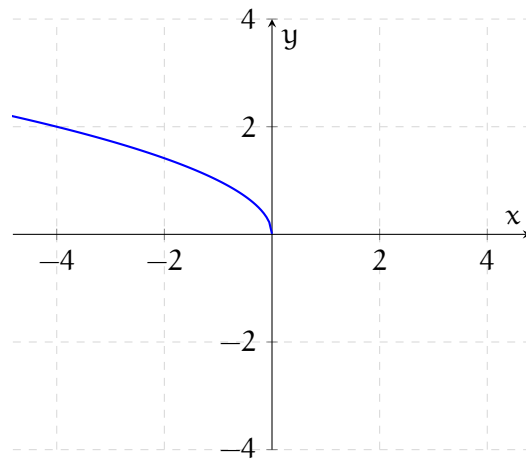


**Definition 2.7 (Reflection About y Axis).**

$$y = f(-x)$$

**Example 2.7 (Reflection About y Axis).**

$$f(x) = \sqrt{-x}$$



## 2.2 How to Graph with Transformations

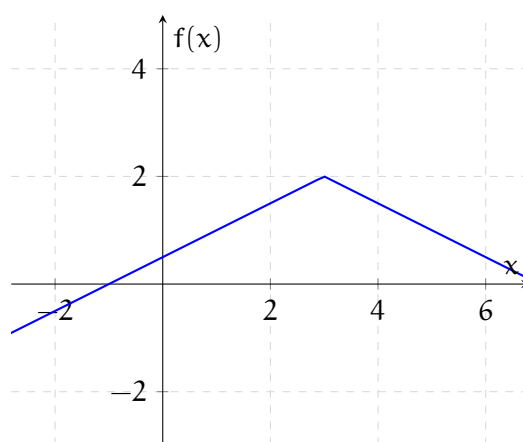
### Example 2.8 (Examples).

1.  $f(x) = -\frac{1}{2}|x - 3| + 2$

Key Points =  $(1, 1)$   $(0, 0)$   $(-1, 1)$

**Steps:**

- (a) Up 2.
- (b) Right 3.
- (c) Vertical Compression.
- (d) Reflect About  $x$ .

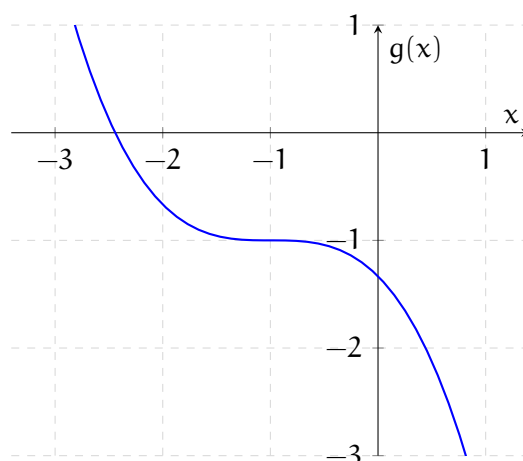


2.  $g(x) = -\frac{1}{3}(x + 1)^3 - 1$

Key Points =  $(1, 1)$   $(0, 0)$   $(-1, -1)$

**Steps:**

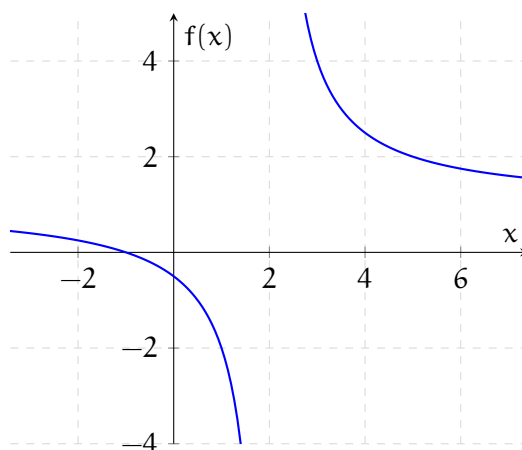
- (a) Down 1.
- (b) Left 1.
- (c) Vertical Compression.
- (d) Reflect about  $x$ .



3.  $f(x) = \frac{3}{x-2} + 1$   
 $f(x) = \frac{3}{x-2} + 1 \rightarrow 3 \cdot \frac{1}{x-2} + 1$   
 Key Points =  $(1, 1)$   $(-1, -1)$

**Steps:**

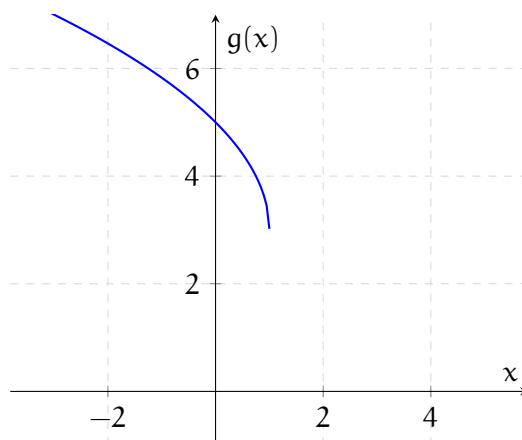
- (a) Up 1.
- (b) Right 2.
- (c) Vertical Stretch.



4.  $g(x) = 2\sqrt{1-x} + 3$   $g(x) = 2\sqrt{1-x} + 3 \rightarrow 2\sqrt{-x+1} + 3$   
 $g(x) = 2\sqrt{-(x-1)} + 3$   
 Key Points =  $(1, 1)$   $(0, 0)$

**Steps:**

- (a) Up 3.
- (b) Right 1.
- (c) Reflect About  $x$ .
- (d) Vertical Stretch.



### 3 QUADRATICS

#### 3.1 Intro to Solving Quadratics

##### Definition 3.1 (Standard Form).

$$f(x) = ax^2 + bx + c$$

$$f(x) = 0$$

$$ax^2 + bx + c = 0$$

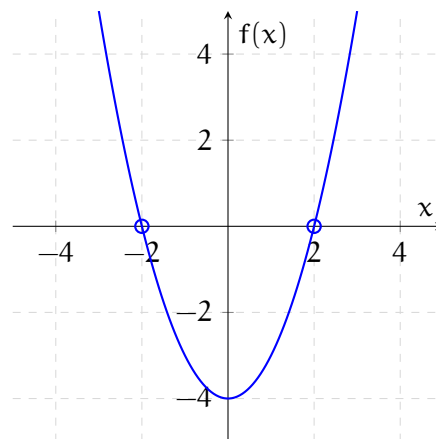
- $a$ ,  $b$  and  $c$  are known numbers (coefficients and constants).
- $a$  cannot be 0, if  $a = 0$ , the equation becomes linear.

**Note:** All we really do is just finding the  $x$  intercepts.

##### 3.1.1 Cases

There are a total of 3 cases a quadratic can have.

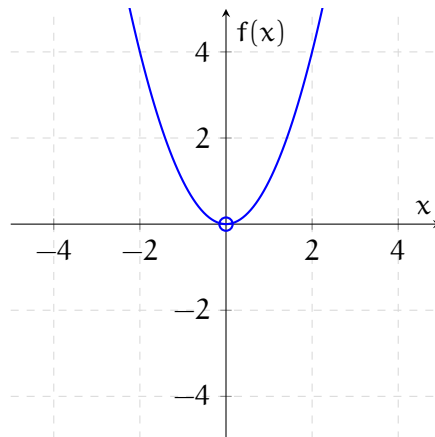
##### Example 3.1 (2 $x$ Intercepts).



- 2 Solutions.
- 2 Real Solutions.
- 2 Real  $x$  Intercepts.

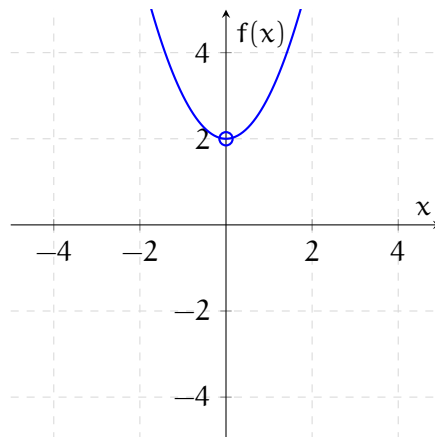
##### Example 3.2 (1 $x$ Intercept).





- 1 Real Solutions.
- 1 Real  $x$  Intercepts.

**Example 3.3 (No  $x$  Intercept).**



- 2 Complex Solutions.
- No  $x$  Intercepts.

### 3.2 The Square Root Method for Quadratics

**Definition 3.2 (Square Root Method).** A technique used to solve quadratic equations by isolating the squared variable term, then taking the square root of both sides.

**Example 3.4 (Square Root Method).**

All of these has 2 solutions.

1.  $f(x) = x^2 - 18$   
 $x^2 - 18 = 0$   
 $x^2 = 18$   
 $\sqrt{x^2} = \pm\sqrt{18}$   
 $x = \pm 3\sqrt{2}$   
 $x = 3\sqrt{2} \wedge x = -3\sqrt{2}$
2.  $f(x) = 3x^2 - 33$   
 $3x^2 - 33 = 0$   
 $3x^2 = 33$   
 $x^2 = 11$   
 $\sqrt{x^2} = \pm\sqrt{11}$   
 $x = \pm\sqrt{11}$   
 $x = \sqrt{11} \wedge x = -\sqrt{11}$
3.  $g(x) = (x + 2)^2 - 1$   
 $(x + 2)^2 - 1 = 0$   
 $(x + 2)^2 = 1$   
 $\sqrt{(x + 2)^2} = \pm\sqrt{1}$   
 $x + 2 = \pm 1$   
 $x = -2 \pm 1$   
 $x = -2 - 1 = -3 \wedge x = -2 + 1 = -1$
4.  $h(x) = (2x + 3)^2 - 32$   
 $(2x + 3)^2 - 32 = 0$   
 $(2x + 3)^2 = 32$   
 $\sqrt{(2x + 3)^2} = \pm\sqrt{32}$   
 $2x + 3 = \pm 4\sqrt{2}$   
 $2x + 3 = \pm 4\sqrt{2}$   
 $2x = -3 \pm 4\sqrt{2}$   
 $x = \frac{-3 \pm 4\sqrt{2}}{2}$   
 $x = \frac{-3 - 4\sqrt{2}}{2} \wedge x = \frac{-3 + 4\sqrt{2}}{2}$

**Definition 3.3 (Imaginary Numbers).** An imaginary number is any number that can be written as a real number multiplied by the imaginary unit  $i$  where  $i = \sqrt{-1}$ .

**Example 3.5 (Special Cases).**

1. **1 Solution.**  
 $f(x) = (x - 7)^2$   
 $(x - 7)^2 = 0$   
 $\sqrt{(x - 7)^2} = \pm\sqrt{0}$   
 $x - 7 = \pm 0$   
 $x = 7 \pm 0$   
 $x = 7 - 0 = 7 \wedge x = 7 + 0 = 7$

**2. No solution (Imaginary Number).**

$$f(x) = (3x - 2)^2 + 75$$

$$(3x - 2)^2 + 75 = 0$$

$$(3x - 2)^2 = -75$$

$$\sqrt{(3x - 2)^2} = \pm \sqrt{-75}$$

$$3x - 2 = \pm 5i\sqrt{3}$$

$$x = \frac{2 \pm 5i\sqrt{3}}{3}$$

$$x = \frac{2+5i\sqrt{3}}{3} \wedge x = \frac{2-5i\sqrt{3}}{3}$$

Can also be written as

$$x = \frac{2}{3} + \frac{5\sqrt{3}}{3}i \wedge x = \frac{2}{3} - \frac{5\sqrt{3}}{3}i$$

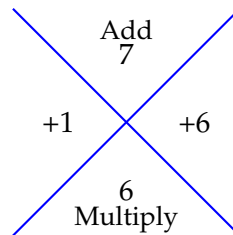
### 3.3 Factoring Method for Quadratics

**Definition 3.4 (Factoring).** Factoring a quadratic is the process of rewriting a standard quadratic expression  $ax^2 + bx + c$  as the product of its linear factors, like  $(x - p)(x - 1)$ .

**Example 3.6 (Factoring).**

$$1. \quad f(x) = x^2 + 7x + 6$$

$$x^2 + 7x + 6 = 0$$



$$x^2 + 1x + 6x + 6 = 0$$

$$x(x + 1) + 6(x + 1) = 0 \rightarrow (x + 1)(x + 6) = 0$$

$$x + 1 = 0 \wedge x + 6 = 0$$

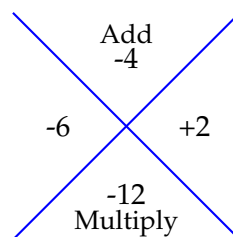
$$x = -1 \wedge x = -6$$

$$2. \quad g(x) = 3x(x - 4) - 36$$

$$3x(x - 4) - 36 = 0$$

$$3x^2 - 12x - 36 = 0$$

$$3(x^2 - 4x - 12) = 0$$



$$3(x^2 - 6x + 2x - 12) = 0$$

$$3[x(x - 6) + 2(x - 6)] = 0$$

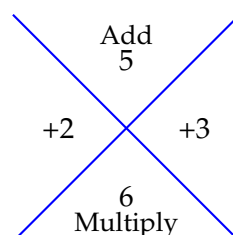
$$3(x - 6)(x + 2) = 0$$

$$x - 6 = 0 \wedge x + 2 = 0$$

$$x = 6 \wedge x = -2$$

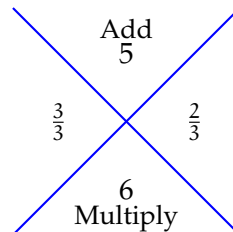
$$3. \quad g(x) = 3x^2 + 2 + 5x$$

$$3x^2 + 5x + 2 = 0$$



$$\begin{aligned}
 3x^2 + 3x + 2x + 2 &= 0 \\
 3x(x+1) + 2(x+1) &= 0 \\
 (x+1)(3x+2) &= 0 \\
 x+1 = 0 \wedge 3x+2 &= 0 \\
 x = -1 \wedge x = -\frac{2}{3}
 \end{aligned}$$

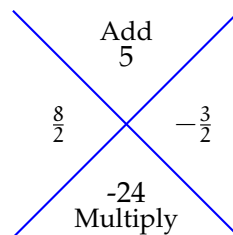
**Note:** The full Shortcut is (We need to divide the factors by a):



4.  $h(x) = -2x^2 - 5x + 12$   
 $-2x^2 - 5x + 12 = 0$

**Note:** We can change the signs here.

$$2x^2 + 5x - 12 = 0$$

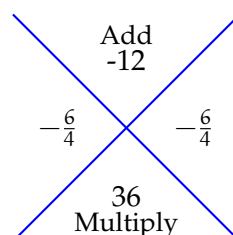


$$\begin{aligned}
 2x^2 + 8x - 3x - 12 &= 0 \\
 2x(x+4) - 3(x+4) &= 0 \\
 (x+4)(2x-3) &= 0 \\
 x+4 = 0 \wedge 2x-3 &= 0 \\
 x = -4 \wedge x = \frac{3}{2}
 \end{aligned}$$

### Example 3.7 (Special Cases).

#### 1. 1 Solution.

$$\begin{aligned}
 f(x) &= -12x + 9 + 4x^2 \\
 4x^2 - 12x + 9 &= 0
 \end{aligned}$$



$$\begin{aligned}
 (2x-3)(2x-3) &= 0 \\
 (2x-3)^2 &= 0
 \end{aligned}$$

**Note:** We can do the square root method here.

$$\sqrt{(2x-3)^2} = \pm\sqrt{0}$$

$$2x-3 = \pm\sqrt{0}$$

$$(x - \frac{3}{2})(x - \frac{3}{2}) = 0$$

$$x = \frac{3}{2} \wedge x = \frac{3}{2}$$

**2. 1 Solution.**

$$g(x) = x^2 - 5$$

$$x^2 - 5 = 0$$

$$x^2 - (\sqrt{5})^2 = 0$$

$$(x - \sqrt{5})(x - \sqrt{5}) = 0$$

$$x = \sqrt{5} \wedge x = \sqrt{5}$$

### 3.4 Completing the Square Method for Quadratics

**Definition 3.5 (Completing the Square).** An algebraic method used to rewrite a quadratic expression from its standard form,  $ax^2 + bx + c$ , into the vertex form,  $a(x - h)^2 + k$ . This transformation creates a perfect square binomial.

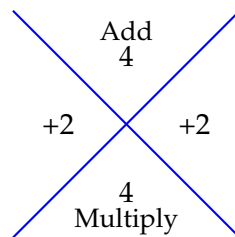
**Example 3.8 (Completing the square).**

$$1. f(x) = x^2 + 4x - 3$$

$$0 = x^2 + 4x - 3$$

**Note:** We're gonna pretend  $-3$  isn't there.

$$0 = x^2 + 4x \quad -3$$



$$0 = (x^2 + 4x + 4) - 4 - 3 \quad (x + 2)(x + 2) = (x + 2)^2$$

$$0 = (x + 2)^2 - 7$$

$$(x + 2)^2 = 7$$

$$\sqrt{(x + 2)^2} = \pm\sqrt{7}$$

$$x + 2 = \pm\sqrt{7}$$

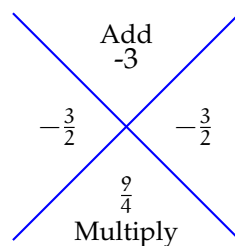
$$x = -2 \pm \sqrt{7}$$

$$x = -2 + \sqrt{7} \wedge x = -2 - \sqrt{7}$$

$$2. g(x) = x^2 - 3x + 1$$

$$x^2 - 3x + 1 = 0$$

$$(x^2 - 3x \quad \quad) + 1 = 0$$



$$(x^2 - 3x + \frac{9}{4}) - \frac{9}{4} + 1 = 0$$

$$(x - \frac{3}{2})^2 - \frac{5}{4} = 0$$

$$(x - \frac{3}{2})^2 = \frac{5}{4}$$

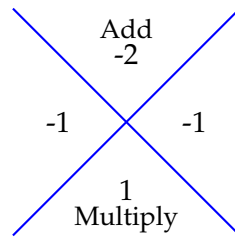
$$\sqrt{(x - \frac{3}{2})^2} = \pm\sqrt{\frac{5}{4}}$$

$$x - \frac{3}{2} = \pm\frac{\sqrt{5}}{2}$$

$$x = \frac{3}{2} \pm \frac{\sqrt{5}}{2}$$

$$x = \frac{3+\sqrt{5}}{2} \wedge x = \frac{3-\sqrt{5}}{2}$$

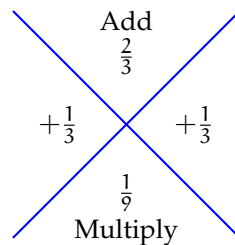
$$\begin{aligned}
 3. \quad h(x) &= 5x^2 - 10x + 2 \\
 5x^2 - 10x + 2 &= 0 \\
 5(x^2 - 2x \quad \quad) + 2 &= 0
 \end{aligned}$$



**Note:** Remember to subtract 1 by 5 because its multiplied by 5.

$$\begin{aligned}
 5(x^2 - 2x + 1) - 5 + 2 &= 0 \\
 5(x - 1)^2 - 3 &= 0 \\
 (x - 1)^2 &= \frac{3}{5} \\
 \sqrt{(x - 1)^2} &= \pm \sqrt{\frac{3}{5}} \\
 x - 1 &= \pm \sqrt{\frac{3}{5}} \\
 x - 1 - \sqrt{\frac{3}{5}} \wedge x - 1 + \sqrt{\frac{3}{5}}
 \end{aligned}$$

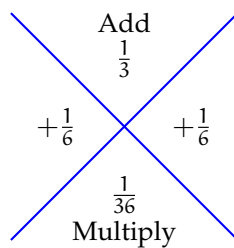
$$\begin{aligned}
 4. \quad g(x) &= x^2 + \frac{2}{3}x - \frac{1}{3} \\
 x^2 &= \frac{2}{3}x - \frac{1}{3} \\
 (x^2 + \frac{2}{3}x \quad \quad) - \frac{1}{3}
 \end{aligned}$$



$$\begin{aligned}
 (x^2 + \frac{2}{3}x + \frac{1}{9}) - \frac{1}{9} - \frac{1}{3} &= 0 \\
 (x + \frac{1}{3})^2 - \frac{4}{9} &= 0 \\
 (x + \frac{1}{3})^2 &= \frac{4}{9} \\
 \sqrt{(x + \frac{1}{3})^2} &= \pm \sqrt{\frac{4}{9}} \\
 x + \frac{1}{3} &= \pm \frac{2}{3} \\
 x &= -\frac{1}{3} \pm \frac{2}{3} \\
 x &= -\frac{1}{3} - \frac{2}{3} \rightarrow \frac{1}{3} \wedge x = -\frac{1}{3} + \frac{2}{3} \rightarrow -1
 \end{aligned}$$

$$\begin{aligned}
 5. \quad f(x) &= 3x^2 + x - \frac{1}{2} \\
 3x^2 + x - \frac{1}{2} &= 0 \\
 3(x^2 + \frac{1}{3}x \quad \quad) - \frac{1}{2} &= 0
 \end{aligned}$$





$$3(x^2 + \frac{1}{3}x + \frac{1}{36}) - \frac{1}{12} - \frac{1}{2} = 0$$

$$3(x + \frac{1}{6})^2 - \frac{7}{12} = 0$$

$$3(x + \frac{1}{6})^2 = \frac{7}{12}$$

$$(x + \frac{1}{6})^2 = \frac{7}{36}$$

$$\sqrt{(x + \frac{1}{6})^2} = \pm \sqrt{\frac{7}{36}}$$

$$x + \frac{1}{6} = \pm \sqrt{\frac{7}{36}}$$

$$x = -\frac{1}{6} \pm \sqrt{\frac{7}{36}}$$

$$x = -\frac{1}{6} + \sqrt{\frac{7}{36}} \wedge x = -\frac{1}{6} - \sqrt{\frac{7}{36}}$$

### 3.5 Proving the Quadratic Formula

#### Definition 3.6 (Quadratic).

$$f(x) = ax^2 + bx + c = 0$$

$$ax^2 + bx + c = 0$$

#### 3.5.1 Proof

$$ax^2 + bx + c = 0$$

$$a\left(x^2 + \frac{b}{a}x\right) + c = 0$$

$$a\left(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}\right) + c = 0$$

$$a\left(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}\right) - \frac{b^2}{4a^2} + c = 0$$

$$a\left(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}\right) - \frac{b^2}{4a^2} + \frac{4ac}{4a}$$

$$a\left(x + \frac{b}{2a}\right)^2 = \frac{b^2}{4a} - \frac{4ac}{4a}$$

$$a\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a}$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

$$\sqrt{\left(x + \frac{b}{2a}\right)^2} = \pm \sqrt{\frac{b^2 - 4ac}{4a^2}}$$

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$x = -b \pm \frac{\sqrt{b^2 - 4ac}}{2a} \quad \square$$

### 3.6 The Quadratic Formula

#### Definition 3.7 (The Quadratic Formula).

For  $ax^2 + bx + c = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The quadratic formula does the same thing as both the square root method and factoring method. The only difference is that it works all the time.

**Note:** Completing the square also works all the time but its harder than the quadratic formula.

#### Example 3.9 (The Quadratic Formula).

1.  $f(x) = 3x^2 - 5x + 1$

$$3x^2 - 5x + 1 = 0$$

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(3)(1)}}{2(3)}$$

$$x = \frac{5 \pm \sqrt{25 - 12}}{6} \rightarrow \frac{5 \pm \sqrt{13}}{6}$$

$$x = \frac{5 + \sqrt{13}}{6} \wedge x = \frac{5 - \sqrt{13}}{6}$$

2.  $g(x) = 3x(x + 2) - 1$

$$3x^2 + 6x - 1 = 0$$

$$x = \frac{-(6) \pm \sqrt{(6)^2 - 4(3)(-1)}}{2(3)}$$

$$x = \frac{-6 \pm \sqrt{36 + 12}}{6}$$

$$x = \frac{-6 \pm \sqrt{48}}{6} \rightarrow \frac{-6 \pm 4\sqrt{3}}{6}$$

$$x = \frac{-3 \pm 2\sqrt{3}}{3}$$

$$x = \frac{-3 + 2\sqrt{3}}{3} \wedge x = \frac{-3 - 2\sqrt{3}}{3}$$

3.  $h(x) = 3x^2 + x - \frac{1}{2}$

$$3x^2 + x - \frac{1}{2} = 0$$

$$2(3x^2 + x - \frac{1}{2}) = 0 \cdot 2$$

$$6x^2 + 2x - 1 = 0$$

$$x = \frac{-(2) \pm \sqrt{(2)^2 - 4(6)(-1)}}{2(6)}$$

$$x = \frac{-2 \pm \sqrt{4 + 24}}{12}$$

$$x = \frac{-2 \pm \sqrt{28}}{12} \rightarrow x = \frac{-2 \pm 2\sqrt{7}}{12}$$

$$x = \frac{-1 \pm \sqrt{7}}{6}$$

$$x = \frac{-1 + \sqrt{7}}{6} \wedge x = \frac{-1 - \sqrt{7}}{6}$$

4.  $f(x) = \frac{1}{8}x^2 - \frac{1}{4}x - 2$

$$\frac{1}{8}x^2 - \frac{1}{4}x - 2 = 0$$

$$8(\frac{1}{8}x^2 - \frac{1}{4}x - 2) = 0 \cdot 8$$

$$x^2 - 2x - 16 = 0$$

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-16)}}{2(1)}$$

$$x = \frac{2 \pm \sqrt{4 + 64}}{2}$$

$$x = \frac{2 \pm \sqrt{68}}{2}$$

$$x = \frac{2 \pm 2\sqrt{17}}{2} \rightarrow 1 \pm \sqrt{17}$$

$$x = 1 + \sqrt{17} \wedge x = 1 - \sqrt{17}$$

**Example 3.10 (Special Cases).**

1.  $f(x) = 9x - 6x + 1$

$$9x - 6x + 1 = 0$$

$$x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(9)(1)}}{2(9)}$$

$$x = \frac{6 \pm \sqrt{36 - 36}}{18}$$

$$x = \frac{6 \pm \sqrt{0}}{18}$$

$$\frac{6+0}{18} \wedge \frac{6-0}{18}$$

$$x = \frac{1}{3}$$

2.  $g(x) = -4x^2 - x - 4$

$$-4x^2 - x - 4 = 0$$

**Note:** We can change all the negatives to positives here.

$$4x^2 + x + 4 = 0$$

$$x = \frac{-(1) \pm \sqrt{(1)^2 - 4(4)(4)}}{2(4)}$$

$$x = \frac{-1 \pm \sqrt{1 - 64}}{8}$$

$$x = \frac{-1 \pm \sqrt{-63}}{8}$$

$$x = \frac{-1 \pm 3i\sqrt{7}}{8}$$

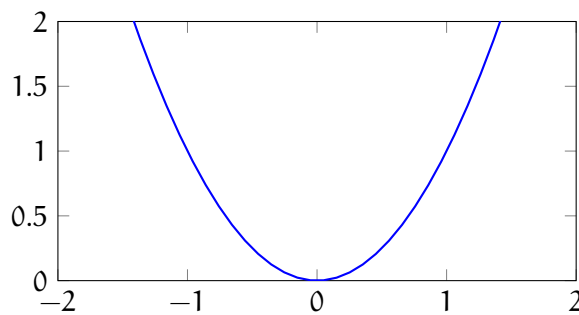
$$x = \frac{-1+3i\sqrt{7}}{8} \wedge \frac{-1-3i\sqrt{7}}{8}$$

### 3.7 Graphs of Quadratics

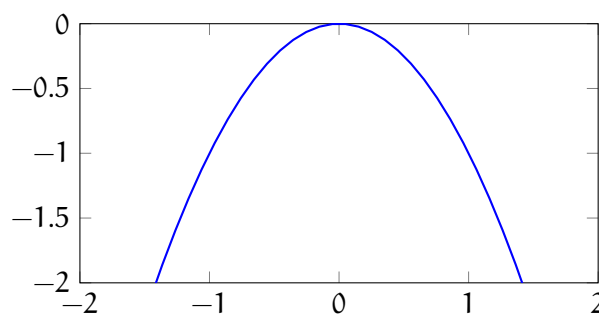
**Definition 3.8 (Graphing Quadratics).** The process of drawing a U-shaped curve called a *parabola* on a grid based on an equation where the highest power of the variable is 2 ( $f(x) = ax^2 + bx + c$ ).

**Note:** The graph will always look like if  $a$  is.

1. if  $a > 0$  it becomes an upward opening parabola.



2. if  $a < 0$  it becomes a downward opening parabola.



**Definition 3.9 (y Intercept).** The point where the graph (a U-shaped curve called a parabola) crosses the vertical  $y$ -axis, always occurring where  $x = 0$ . The  $y$ -intercept is simply the constant number  $c$ .

$$(0, c)$$

**Definition 3.10 (Vertex).** The turning point on a parabola's U-shaped graph where the function reaches its maximum or minimum value, and the graph changes direction.

$$\left( \frac{b}{2a}, f(x) \right)$$

Or we can complete the square for the vertex form.

**Example 3.11 (Quadratics Graphs).**

1.  $f(x) = 2x^2 + 8x + 5$

(a) Vertex:

$$\left( -\frac{8}{2(2)}, \right)$$

**Note:** We plug in  $-2$  into the function.

$$(-2, -3)$$

(b) y-intercept:

$$(0, 5)$$

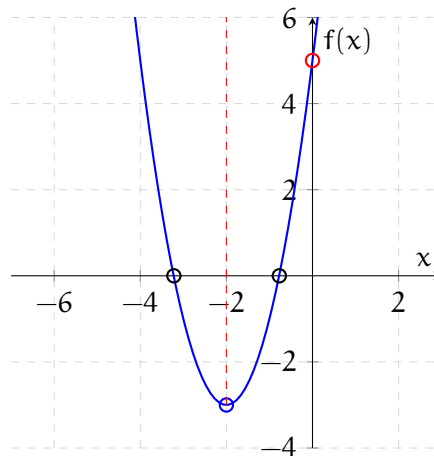
(c) x-intercept:

$$2x^2 + 8x + 5 = 0$$

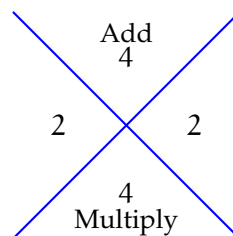
$$x = \frac{-(-8) \pm \sqrt{(-8)^2 - 4(2)(5)}}{2(2)}$$

$$x = \frac{-8 \pm \sqrt{24}}{4}$$

$$x \approx -3.22 \wedge x \approx -0.77$$

**Getting the vertex form:**

$$f(x) = 2(x^2 + 4x \quad ) + 5$$



$$f(x) = 2(x^2 + 4x + 4) - 8 + 5$$

$$2(x + 2)^2 - 3$$

$$f(x) = 2(x + 2)^2 - 3$$

2.  $g(x) = -2x^2 - 4x + 2$

(a) Vertex:

$$\left( -\frac{(-4)}{2(-2)}, \quad \right)$$

$$(-1, 4)$$

(b) y-intercept:

$$(0, 2)$$

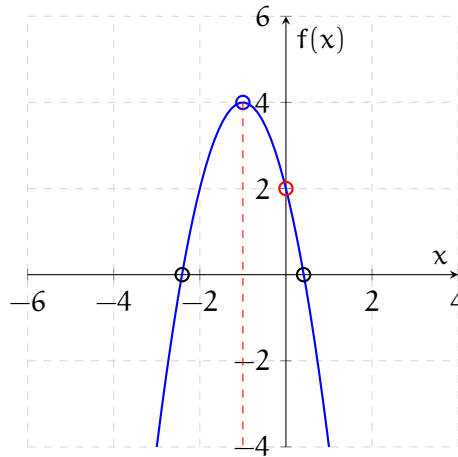
(c) x-intercept:

$$2x^2 - 4x + 2 = 0$$

$$2(x^2 + 2x - 1) = 0$$

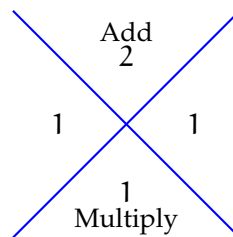
$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-1)}}{2(1)}$$

$$\begin{aligned}
 x &= \frac{-2 \pm \sqrt{8}}{2} \\
 x &= \frac{-2 \pm 2\sqrt{2}}{2} \\
 x &= -1 \pm \sqrt{2} \\
 x &\approx 0.41 \wedge x \approx -2.41
 \end{aligned}$$



**Getting the vertex form:**

$$g(x) = -2(x^2 + 2x \quad ) + 2$$



$$\begin{aligned}
 g(x) &= -2(x^2 + 2x + 1) + 2 + 2 \\
 g(x) &= -2(x + 1)^2 + 4
 \end{aligned}$$

3.  $f(x) = -4x^2 - 4x - 1$

(a) Vertex:

$$\begin{aligned}
 &\left( -\frac{(-4)}{2(-4)}, \quad \right) \\
 &\left( -\frac{1}{2}, 0 \right)
 \end{aligned}$$

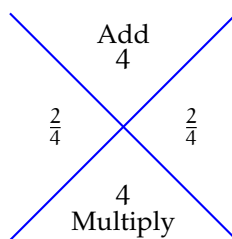
(b) y-intercept:

$$(0, -1)$$

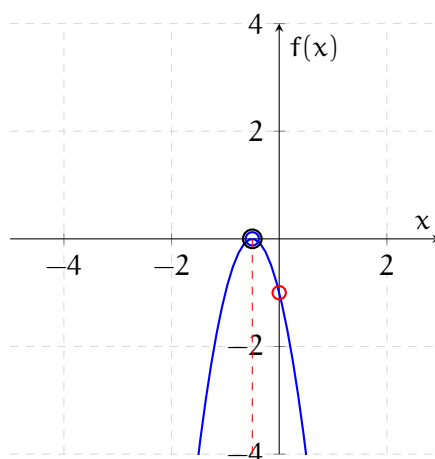
(c) x-intercept:

$$-4x^2 - 4x - 1 = 0$$

$$4x^2 + 4x + 1 = 0$$



$$x = -\frac{1}{2}$$



4.  $g(x) = 3x^2 + 2x + 5$

(a) Vertex:

$$\left(-\frac{(2)}{2(3)}, \quad \right)$$

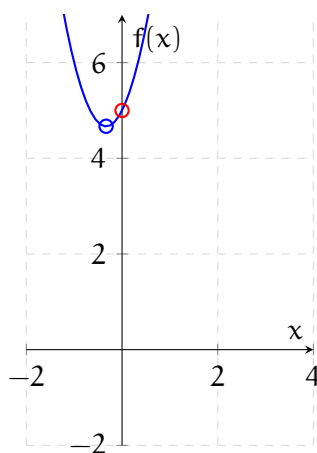
$$\left(-\frac{1}{3}, 4\frac{2}{3}\right)$$

(b) y-intercept:

$$(0, 5)$$

(c) x-intercept:

None.





### 3.8 Quadratics Inequalities

**Definition 3.11 (Quadratics Inequalities).** A math sentence with a squared term where the highest power is 2, and it uses signs like  $>$ ,  $<$ ,  $\geq$ , or  $\leq$  instead of a  $=$  sign.

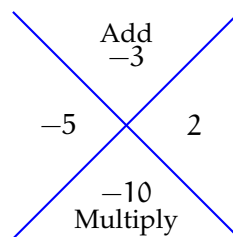
The standard forms is as follows:

$$a \neq 0$$

- $ax^2 + bx + c > 0$
- $ax^2 + bx + c < 0$
- $ax^2 + bx + c \geq 0$
- $ax^2 + bx + c \leq 0$

**Example 3.12 (Quadratics Inequalities).**

$$1. \quad \begin{aligned} x^2 - 3x - 10 &\leq 0 \\ x^2 - 3x - 10 &= 0 \end{aligned}$$

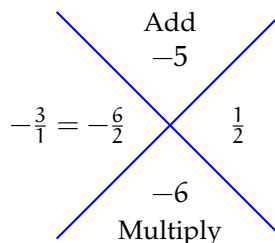


$$\begin{aligned} (x-5)(x+2) &= 0 \\ x-5 &= 0 \wedge x+2 = 0 \\ x &= 5 \wedge x = -2 \\ [-2, 5] \\ (0)^2 - 3(0) - 10 &\leq 0 \\ v(-10 \leq 0) &= \top \end{aligned}$$

$$2. \quad 2x^2 < 5x + 3$$

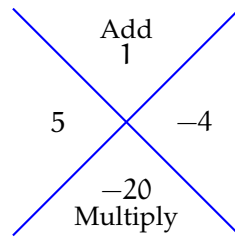
**Note:** We can actually change the inequalities and still get the same answer.

$$2x^2 - 5x - 3 < 0$$



$$\begin{aligned} x &= 3 \wedge x = -\frac{1}{2} \\ (-\frac{1}{2}, 3) \\ 2(0) - 5(0) - 3 &< 0 \\ v(-3 < 0) &= \top \end{aligned}$$

$$\begin{aligned}
 3. \quad & x(x+1) > 20 \\
 & x^2 + x > 20 \\
 & x^2 + x - 20 > 0
 \end{aligned}$$



$$\begin{aligned}
 & (x+5)(x-4) > 0 \\
 & x = -5 \wedge x = 4 \\
 & (-\infty, -5) \cup (4, \infty) \\
 & (0)^2 + (0) - 20 > 0 \\
 & v(-20 > 0) = \perp
 \end{aligned}$$

$$\begin{aligned}
 4. \quad & 4x^2 + 9 < 6x \\
 & 4x^2 - 6x + 9 < 0 \\
 & x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(4)(9)}}{2(4)} \\
 & x = \frac{6 \pm \sqrt{36 - 144}}{8}
 \end{aligned}$$

**Note:** This parabola has **NO** x-axis.

$$\begin{aligned}
 & 4(0)^2 - 6(0) + 9 < 0 \\
 & v(9 < 0) = \perp
 \end{aligned}$$

## 4 INTERSECTION OF FUNCTIONS

**Definition 4.1 (Intersection of Functions).** The point or set of points where both functions share the exact same input and output values.

**Example 4.1 (Intersection of Functions).**

$$1. f(x) = x^2 + 6x + 3 \quad g(x) = 3$$

$$x^2 + 6x + 3 = 3$$

$$x^2 + 6x + 3 - 3 = 0$$

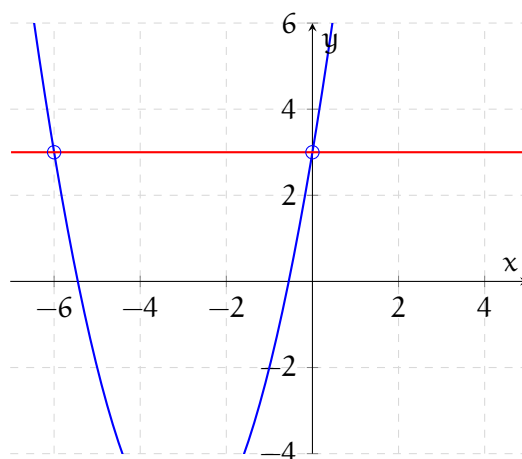
$$x^2 + 6x = 0$$

$$x(x + 6) = 0$$

$$x = 0, x + 6 = 0$$

$$x = 0, x = -6$$

$$(0, 3), (-6, 3)$$

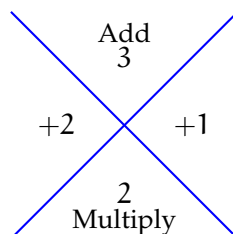


$$2. -2x^2 + 1 \quad g(x) = 3x + 2$$

$$-2x^2 + 1 = 3x + 2$$

$$0 = 2x^2 + 3x - 1 + 2$$

$$0 = 2x^2 + 3x + 1$$

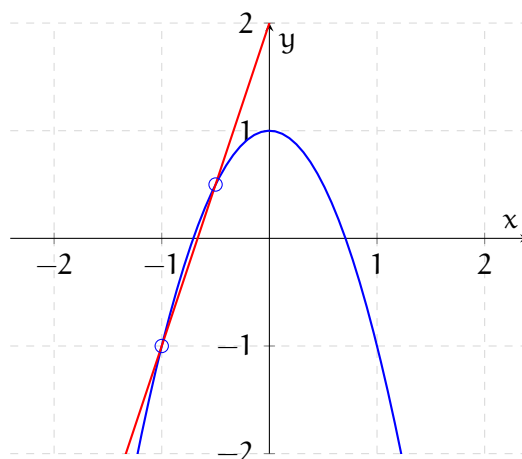


$$0 = (x + 1)(2x + 1)$$

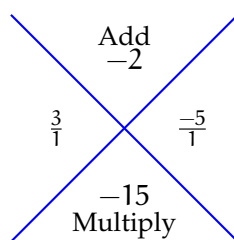
$$x + 1 = 0, 2x + 1 = 0$$

$$x = -1, x = -\frac{1}{2}$$

$$(-1, -1), \left(-\frac{1}{2}, \frac{1}{2}\right)$$

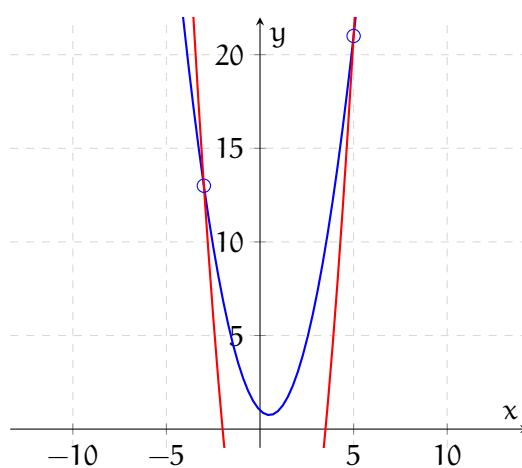


$$\begin{aligned}
 3. \quad f(x) &= x^2 - x + 1 \quad g(x) = 2x^2 - 3x - 14 \\
 x^2 - x + 1 &= 2x^2 - 3x - 14 \\
 0 &= -x^2 + 2x^2 + x - 3x - 1 - 14 \\
 0 &= x^2 - 2x - 15
 \end{aligned}$$



$$\begin{aligned}
 0 &= (x + 3)(x - 5) \\
 x + 3 &= 0, x - 5 = 0 \\
 x &= -3, x = 5
 \end{aligned}$$

$$(-3, 13)(5, 21)$$



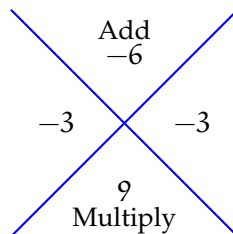
## 5 SOLVING EQUATIONS BY USING SUBSTITUTIONS

**Note:** also called u-substitution.

**Definition 5.1 (u-substitution).** Solves equations of quadratic form by rewriting them as  $a(g(x))^2 + b(g(x)) + c = 0$  where  $u = g(x)$  transforms the equation into the standard quadratic form  $au^2 + bu + c = 0$ .

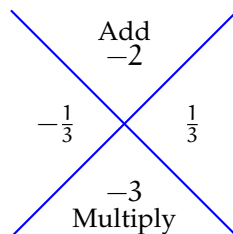
### Example 5.1 (Equations).

$$\begin{aligned} 1. \quad & g(x) = (3x + 4)^2 - 6(3x + 4) + 9 \\ & (3x + 4)^2 - 6(3x + 4) + 9 = 0 \\ & u = 3x + 4 \\ & u^2 - 6u + 9 = 0 \end{aligned}$$



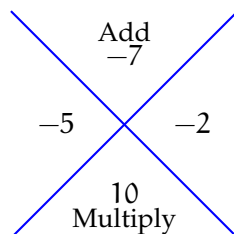
$$\begin{aligned} & (u - 3)(u - 3) = 0 \\ & (u - 3)^2 = 0 \\ & (u - 3) = 0 \\ & u = 3 \rightarrow 3x + 4 = 3 \rightarrow 3x + 4 = 3 \\ & 3x = -1 \\ & x = -\frac{1}{3} \end{aligned}$$

$$\begin{aligned} 2. \quad & f(x) = 3x^4 - 2x^2 - 1 \\ & 3x^4 - 2x^2 - 1 = 0 \\ & 3(x^2)^2 - 2(x^2) - 1 = 0 \\ & u = x^2 \\ & 3u^2 - 2u - 1 = 0 \end{aligned}$$



$$\begin{aligned} & u = 1 \wedge u = -\frac{1}{3} \\ & (u - 1)(3u + 1) = 0 \\ & u - 1 = 0 \wedge 3u + 1 = 0 \\ & x^2 = 1 \wedge x^2 = -\frac{1}{3} \\ & \sqrt{x^2} = \pm\sqrt{1} \wedge \sqrt{x^2} = \pm\sqrt{-\frac{1}{3}} \\ & (x = 1 \wedge x = i\sqrt{\frac{1}{3}}) \wedge (x = -1 \wedge x = -i\sqrt{\frac{1}{3}}) \end{aligned}$$

$$\begin{aligned}
 3. \quad & f(x) = x^{\frac{2}{3}} - 7x^{\frac{1}{3}} + 10 \\
 & x^{\frac{2}{3}} - 7x^{\frac{1}{3}} + 10 = 0 \\
 & (x^{\frac{1}{3}})^2 - 7(x^{\frac{1}{3}}) + 10 = 0 \\
 & u = x^{\frac{1}{3}} \\
 & u^2 - 7u + 10 = 0
 \end{aligned}$$

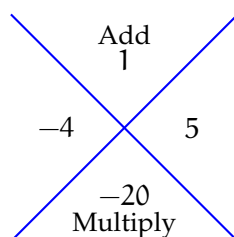


$$\begin{aligned}
 (u - 5)(u - 2) &= 0 \\
 u - 5 = 0 \wedge u - 2 &= 0 \\
 u = 5 \wedge u &= 2 \\
 x^{\frac{1}{3}} = 5 \wedge x^{\frac{1}{3}} &= 2
 \end{aligned}$$

**Note:** We can do two methods here.

$$\begin{aligned}
 \sqrt[3]{x} &= 5 \\
 (\sqrt[3]{x})^3 &= (5)^3 \\
 x &= 125 \\
 x^{\frac{1}{3}} &= 2 \\
 (x^{\frac{1}{3}})^3 &= (2)^3 \\
 x &= 8
 \end{aligned}$$

$$\begin{aligned}
 4. \quad & g(x) = x + \sqrt{x} - 20 \\
 & x + \sqrt{x^2} - 20 = 0 \\
 & x^1 + x^{\frac{1}{2}} - 20 = 0 \\
 & (x^{\frac{1}{2}})^2 + (x^{\frac{1}{2}}) - 20 = 0 \\
 & u = x^{\frac{1}{2}} \\
 & u^2 + u - 20 = 0
 \end{aligned}$$



$$\begin{aligned}
 u = 4 \wedge u &= -5 \\
 x^{\frac{1}{2}} = 4 \wedge x^{\frac{1}{2}} &= -5 \\
 \sqrt{x} = 4 \wedge \sqrt{x} &= -5 \\
 (\sqrt{x})^2 &= (4)^2 \wedge (\sqrt{x})^2 = (-5)^2 \\
 x = 16 \wedge x &= 25
 \end{aligned}$$

**Note:**  $x = 25$  is not a real solution because a square root can't equal a negative.

$$\begin{aligned}
 5. \quad & h(x) = -2y^{-2} - 11y^{-1} + 40 \\
 & -2y^{-2} - 11y^{-1} + 40 = 0 \\
 & 2y^{-2} + 11y^{-1} - 40 = 0
 \end{aligned}$$

$$2(y^{-1})^2 + 11(y^{-1}) - 40 = 0$$

$$u = y^{-1}$$

$$2u^2 + 11u - 40 = 0$$

Add  
11

$$-\frac{5}{2} = \frac{16}{2} = \frac{8}{1}$$

-80  
Multiply

$$(2u - 5)(u + 8) = 0$$

$$2u - 5 = 0 \wedge u + 8 = 0$$

$$u = \frac{5}{2} \wedge u = -8$$

$$y^{-1} = \frac{5}{2} \wedge y^{-1} = -8$$

$$\frac{1}{y} = \frac{5}{2} \wedge \frac{1}{y} = -8$$

$$y = \frac{2}{5} \wedge y = -\frac{1}{8}$$

## 6 POLYNOMIALS

### 6.1 Introduction to Polynomials

**Definition 6.1 (Polynomial Function).** A polynomial function, in general, is also stated as a polynomial or polynomial expression, defined by its degree. The degree of any polynomial is the highest power present in it.

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

Leading Term:  $a_n x^n$

Constant Term:  $a_0$

Degree:  $n$  (*Largest exponent in the polynomial.*)

The domain of a polynomial is all  $\mathbb{R}$  numbers.

**Note:** Exponents are all non negative integers.

#### Example 6.1 (Polynomial Function).

1.  $f(x) = 5x^2 + 4x^4$

$$f(x) = 4x^4 + 5x^2$$

This is a degree 4 polynomial.

2.  $g(x) = 3 - \frac{1}{2}x$

$$-\frac{1}{2}x + 3$$

This is a degree 1 polynomial.

3.  $h(x) = 9$

This is a degree 0 polynomial.

4.  $f(x) = 0$

This function doesn't have a degree.

5.  $f(x) = 9x^4 - \pi x^2 + \frac{1}{2}$

6.  $h(x) = \frac{3x^2-1}{7}$

$$\frac{3x^2}{7} - \frac{1}{7}$$

$$\frac{3}{7}x^2 - \frac{1}{7}$$

7.  $F(x) = 2x^3(x^2 - 1)(x - 1)^2$

**Note:** We can do a fake distribution here.

$$2 \cdot x^3 \cdot (x^2 - 1)^1 \cdot (x^1 - 1)^2$$

$$2 \cdot x^3 \cdot x^2 \cdot x^2$$

$$2x^7$$

#### Example 6.2 (Non-Polynomial Function).

1.  $F(x) = x^{\frac{2}{3}} + 2x - 1$

2.  $G(x) = 1 - \frac{4}{x}$



$$3. \ g(x) = \frac{x^2-5}{x^4}$$
$$\frac{x^2}{x^4} - \frac{5}{x^4}$$
$$\frac{1}{x^2} - \frac{5}{x^4}$$

$$4. \ G(x) = 2\sqrt{x}(\sqrt{x} + 1)$$